ENGINEERING ECONOMY

Nominal and effective interest rates and continuous compounding

INTRO

- Since many real world problems involve payments and compounding periods which are not equal to one year, it is necessary to understand nominal and effective interest rates.

- If the compounding period is made infinitely small, interest is compounded continuously.

- All the engineering economy formulas require the use of effective interest rates only.

INTRO

- In some instances there are several effective rates that can be used to solve the problem (such as when only single-payment amounts are involved).

- In other cases, only one effective rate can be used (Uniform series problems)

- For uniform series problems, the interest and payment periods must agree. If the compounding period is shorter than the payment period, then the interest rate must be manipulated to obtain an effective rate over the payment period.

- When the compounding period is longer than the payment period, the payments are manipulated so that the cash flows coincide with compounding periods (deposits are moved to the end and withdrawals are moved to the beginning of the period).

SIMPLE AND COMPOUND INTEREST

- The concepts of simple and compound interests have been introduced.
- The basic difference between the two is that compound interest includes interest on interest earned in the previous period while simple interest does not.
- In essence nominal and effective interest rates have the same relationship to each other as do simple and compound interest.
- The difference is that nominal and effective interests are used when the compounding period (or interest period) is less than one year. Thus when an interest rate is expressed over a period of time shorter than a year, such as 1% per month, the terms nominal and effective interest rates must be considered.

NOMINAL INTEREST RATE

- A dictionary definition of the word nominal is “purported, so-called, ostensible, or professed.” These synonyms imply that a nominal interest rate is not a correct, genuine or effective interest rate. Nominal interest rates must be converted into effective interest rates in order to accurately reflect time-value considerations.

- Nominal interest rate, r, can be defined as the period interest rate times the number of periods.

- r = interest rate per period x number of periods

- A nominal interest can be found for any time period which is longer than the originally stated period.

- eg. 1.5% per month could also be expressed as a nominal 4.5% per quarter (that is 1.5% per month x 3 months per quarter), 9.0% per semiannual period, 18% per year, 36% per two years, etc.

NOMINAL INTEREST RATE

- Nominal interest rate obviously ignores the time value of money and the frequency with which interest is compounded. When the time value of money is taken into consideration in calculating interest rates from period interest rates, the rate is called an effective interest rate.
- Effective rates can also be determined for any period longer than the originally stated period.
- **Note – All formulas derived are based on compound interest, and therefore, only effective interest rates can be used in the equations.**

Interest rates can be expressed or stated in different ways. The table below shows the three general ways in which interest rates can be stated.

Interest Rate Statement	**Interpretation**
• i = 12% per year	• i = effective 12% per year compounded yearly
• i = 1% per month	• i = effective 1% per month compounded monthly
• i = 3½% per quarter	• i = effective 3½ per quarter compounded quarterly

- Comment
- When no compounding period is given, interest rate is an effective rate, with compounding period assumed to be equal to stated time period

Interest rate statements and their interpretation

Interest Rate Statement	Interpretation
• i = 8% per year, compounded monthly	• i = nominal 8% per year compounded monthl
• i = 4% per quarter compounded monthly	• i = nominal 4% per quarter compounded monthly
• i = 14% per year compounded semiannually	• i = nominal 14% per year compounded semiannually

- When compounding period is given without stating whether the interest rate is nominal or effective, it is assumed to be nominal. Compounding period is as stated.
- *i* = effective 10% per year compounded monthly
- *i* = effective 6% per quarter
-
- *i* = effective 1% per month compounded daily

- *i* = effective 10% per year compounded monthly
- *i* = effective 6% per quarter compounded quarterly
- *i* = effective 1% per month compounded daily
- If interest rate is stated as an effective rate, then it is an effective rate. If compounding period is not given, compounding period is assumed to coincide with stated time period.

Effective interest rate formulation

- To illustrate the difference between nominal and effective interest rate, the future worth of $100 after 1 year is determined using both rates. If the bank pays 12% interest compounded annually, the future worth of $100 is given as

- $F = P(1 = i)^n = 100(1.12)^1 = \$112.00$

Effective interest rate formulation

- On the other hand, if the bank pays interest that is compounded semiannually, the future worth must include the interest in the interest earned in the first period.
- An interest rate of 12% per year compounded semiannually means that the bank will pay 6% interest after 6 months and another 6% after 12 months (i.e., every 6 months).

Effective interest rate formulation

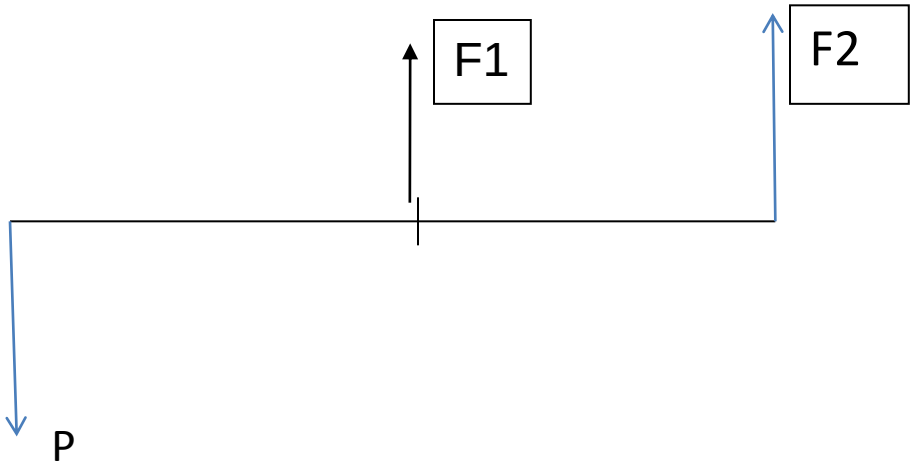
- The first calculation ignores the interest earned in the first 6-month period. The future worth values of $100 after 6 months and after 12 months are:

- $F_6 = 100(1 + 0.06)^1 = \$106.00$

- $F_{12} = 106(1 + 0.06)^1 = \$112.36$

- where 6% is the effective semiannual interest rate. The interest earned in 1 year is $12.36 instead of $12.00, therefore effective annual interest rate is 12.36%.

Cash-flow Diagram for semi annual compounding



Determination of effective interest rate from the nominal interest rate

- The equation may be generalized as follows:

$$i = \left(1 + \frac{r}{m}\right)^{m} - 1$$

- Where *i* = effective interest rate per period
- *r* = nominal interest rate per period
- *m* = number of compounding periods

- A national credit card carries an interest rate of 2% per month on an unpaid balance.
- a) Calculate the effective rate per semiannual period.
- b) If the interest rate is stated as 5% per quarter find the effective rates per semiannual and annual time periods.

- r = 2% per month x 6 months per semiannual period
- = 12% per semiannual period

- The *m* in the effective interest rate equation is equal to 6 since interest would be compounded six times in a 6-months time period.

  i per 6 months is given as:

$$i = \left(1 + \frac{0.12}{6}\right)^{6} - 1$$

$$= 0.1262 \qquad (12.62\%)$$

- For an interest rate of 5% per quarter, the compounding period is quarterly. Therefore, in a semiannual period, $m = 2$ and $r = 10\%$. Thus,
- i per 6 months is given as:

$$i = \left(1 + \frac{0.10}{2}\right)^2 - 1$$

$$= \quad 0.1025 \qquad (10.25\%)$$

- The effective interest rate per year can be determined using r = 20% and m = 4, as follows:

$$i = \left(1 + \frac{0.20}{4}\right)^4 - 1$$

- = 0.2155 (21.55%)

- Note that the term r/m in the effective interest rate equation is always equal to the interest rate (effective) per compounding period. In part a) it was 2% per month while in part b) it was 5% per quarter.